

Symbolic Retrieval

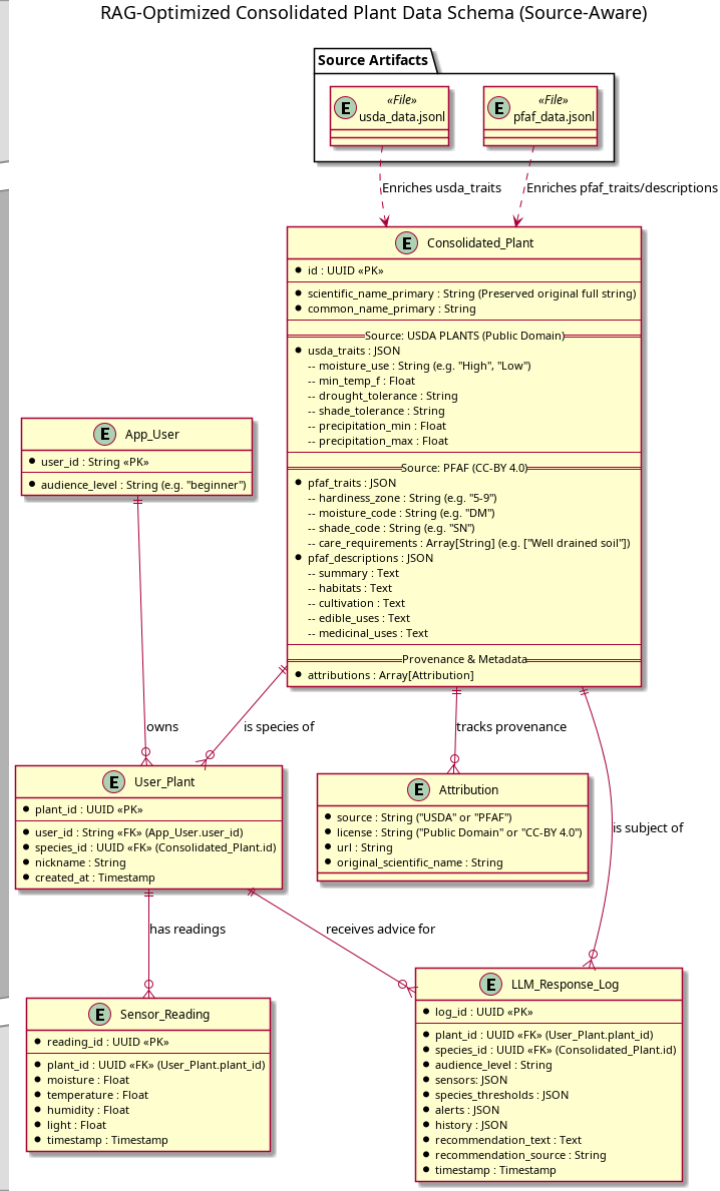
- Refresh: Symbolic Retrieval is our purposed grounding method.
- We give the AI Data, which it in turn uses to generate a response.
- To do this we needed a database to make our comparisons.

The Database

- Scraped data from the internet.
- Used USDA Plants database (Public Domain) and PFAF database (CC 4.0)
- Converted both to JSON format.
- Merged them by Genius-Species names.
- Filtered out entries without light, soil, and moisture metrics.
- 11,383 plants.

Data Structure

- A fully queryable data structure.
- Used SQLite3 and Python.
 - (It's SQL with extra steps.)
- Supports logging of LLM responses.
- Supports user experience level.



The API

- PlantDatabase class with methods for storage and retrieval.
- Supports querying the database for a specific species.
- CRUD operations for user plants.
- Storage and retrieval of sensor data.
- Full logging support.
- The *so, what?*:
 - Allows us to supply the back-end with a wealth of data.
 - Supports individual users and their plants.
 - Logs all LLM interactions for safety and debugging.